Y-FFC Net for 3D Reconstruction of Highly Reflective Surfaces

Xuwen Song and Lianpo Wang

***Abstract*—3D reconstruction plays a pivotal role in intelligent manufacturing, industrial inspection, and other fields. Fringe projection profilometry is a widely used 3D reconstruction method due to its high accuracy and noncontact nature. However, when fringe projection profilometry (FPP) is applied to reconstruct highly reflective surfaces, there are not only diffuse reflections, but also undesired specular reflections. Intense specular reflection destroys the sinusoidal characteristics of the fringe pattern, leading to a decrease in 3D reconstruction accuracy. Traditional methods, such as Multiple exposures method, require multiple redundant shots to obtain high-quality fringe patterns, which makes efficiency difficult to meet industrial measurement requirements. Inspired by the application of deep learning in 3D vision, this article proposes a Y-shaped fast Fourier convolutional network (Y-FFC) for high reflection removal to provide high-quality fringe patterns. The cosine characteristics of the grayscale gradient of the sinusoidal fringe pattern make the detection of nonsinusoidal regions more accurate in the gradient domain. In addition, the undesired specular reflection component is easier to distinguish and filter out in the frequency domain. Therefore, the design of Y-FFC fully considers the gradient and frequency information. Experimental results show that the introduction of gradient and frequency information is beneficial for high reflection removal, enabling the reconstruction of industrial workpieces with highly reflective surfaces without the need for additional shots. The mean absolute error for depth measurements of an aircraft blade with a depth of 30 mm was reduced from 0.1332 to 0.0041 mm.**

***Index Terms*—Fast Fourier convolutional (FFC), fringe projection profilometry, high reflection removal, industrial metrology.**

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The dataset and network code are available at LianpoWang/Y-FFC (github.com).

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I. Introduction

3D reconstruction techniques are widely used in industrial fields such as automotive manufacturing [1] and aerospace engineering [2], and can be divided into contact methods (e.g., laser tracker and coordinate measuring machine) and noncontact methods (e.g., stereo vision and structured light). Among these 3D reconstruction methods, fringe projection profilometry (FPP) stands out in industrial fields due to its high accuracy, easy-to-operate, low environmental requirements, and noninvasive characteristics. The FPP system consists of a projector and a camera, where the former illuminates the object with pre-coded sinusoidal fringe patterns [3], and the latter simultaneously captures deformed fringe patterns from which the 3D shape of the object can be calculated [4]. However, when traditional FPP is applied to industrial scenes with highly reflective objects (e.g., metals, electroplated surfaces, etc.), the reconstructed 3D shapes exhibit flaws and fluctuations. This is due to the presence of two types of reflections on highly reflective surfaces: diffuse and specular reflections. Specular reflection tends to produce overexposed saturated regions, and even in nonoverexposed regions, the fringe sinusoidal characteristics can also be seriously destroyed. FPP cannot accurately calculate the 3D shape from nonsinusoidal fringes, so removing high reflections from the fringe patterns is the foundation for subsequent high-precision FPP reconstruction.

To mitigate the curse of high reflection of fringe patterns, traditional methods typically can be divided into two categories: hardware-based and algorithm-based methods. Hardware based methods include adjusting system hardware parameters (e.g., Multiple exposures method [5]) and adding optical accessories. However, hardware-based methods are time-consuming, expensive, and difficult to control. Algorithm-based methods mainly include increasing the number of phase-shift steps [6] and using inverse fringe methods [7]. However, these methods also require multiple redundant shots and are less efficient.

In recent years, convolutional neural networks have been widely used in high reflection removal tasks [8], and gradually extended to the FPP field. Yu et al. [9] and Liu et al. [10] solved the overexposure problem by setting a lower exposure time to compress the grayscale amplitude. Although they subsequently stretched the grayscale amplitude through neural networks, the accuracy will be greatly reduced when the modulation is severely compressed. To avoid modulation compression, Li et al. [11] solved the high reflection problem at normal exposure times by detecting overexposed regions using a two-channel U-Net

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network and enhancing them with a dilated residual network. But this method ignored the sinusoidal information loss caused by specular reflections in other nonoverexposed regions. To address the problem of simultaneously existing overexposed saturated areas and underexposed dark areas in fringe patterns caused by uneven surface reflectivity, Zhang et al. [12] and Ueda et al. [13], respectively, used CNN and residual UNet to compress saturated fringes and enhance dark fringes. But both methods required the input of multiple fringe patterns with different phase shifts, and subsequent phase retrieval was limited to using a fixed-step phase shift algorithm. Yang et al. [14] improved these methods by inputting a single fringe pattern, using a two-stage network to first segment saturated and dark regions, and then enhancing these regions. However, these methods only focus on brightness value, whereas the factor that affects the reconstruction accuracy is actually the sinusoidal characteristics of the fringe, which may also be poor in normal brightness regions.

To simultaneously address overexposure problems and non-sinusoidal fringe problems caused by high reflection, this article proposes a Y-shaped fast Fourier convolution network (Y-FFC) for removing high reflections. The input highly reflective fringe pattern is separated into a diffuse reflection component with lossless sinusoidal fringes and an irrelevant specular reflection component after passing through the Y-FFC network. Y-FFC consists of an encoder and a dual-branch decoder, where the encoder is used to extract contextual features. The main decoder is used to separate diffuse and specular reflection components, which incorporates fast Fourier convolution (FFC) [17] to simplify the separation task by utilizing frequency domain information. The other decoder is the gradient decoder, which uses multiscale gradient information to identify nonsinusoidal regions, helping the main decoder to give different attention to different regions affected by high reflection. It is worth noting that the Y-shaped UNet with shared encoder has been widely used in the optical field, such as image segmentation [15], absolute phase retrieval in FPP [16], and so on. Our contributions are as follows.

1) Fringe sinusoidal characteristics are utilized in both the frequency and gradient domains. The main decoder removes undesired frequencies introduced by specular reflections from the frequency domain. The gradient decoder can accurately identify nonsinusoidal regions.
2) Diverse loss functions are designed considering the intrinsic relationship between diffuse and specular reflections. With the participation of specular reflection in the loss function calculation, the detection accuracy of the diffuse reflection component can be improved by flexibly changing the weight of specular reflection.
3) The training dataset is generated by utilizing the virtual software 3Dss Max, which solves the efficiency bottleneck of acquiring large-scale data, and at the same time ensures the accuracy and authenticity of the dataset.

II. Basic Principle

A. FPP Principle

Fig. 1(a) shows a typical setup of the FPP hardware system which consists of a camera and a projector. The FPP process is shown in Fig. 1(b). First, the projector projects multiple fringe patterns with different frequencies and phase shifts onto the object, and the camera captures the deformed fringe patterns modulated by depth information, which can be represented as

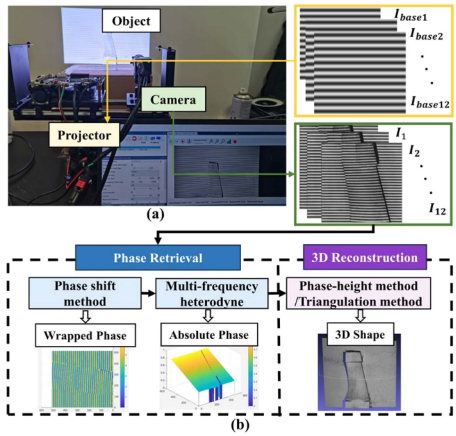


Fig. 1. (a) System setup of FPP. (b) 3D reconstruction using FPP.

$$I_i(x,y) = A(x,y) + B(x,y)\cos[\varphi(x,y) + \delta_i], i = 1, 2 \ldots N \tag{1}$$

where A and B are the background and modulated light intensities, and φ is wrapped phase. δ_i is the phase shift, and N is the phase shift steps. The wrapped phase is usually calculated using phase shift algorithm [18], among which the four-step phase shift (four-step PS) algorithm is widely used due to its balance of accuracy and speed. Since the wrapped phase is folded from 0 to 2π, phase unwrapping is required to obtain the unique absolute phase in the entire image. Temporal phase unwrapping methods, such as the three-frequency heterodyne method [19] is widely used as it can handle complex cases such as geometrically discontinuous objects. Finally, the depth map can be reconstructed using the phase height method by calibrating the phase height coefficients. The point cloud can be reconstructed using triangulation by establishing the geometric relationship between the projector, camera, and object [4].

B. Reconstruction Error Caused by High Reflection

High-precision FPP generally requires the object to be a diffuse reflective surface. When the object is a highly reflective surface, there will be both specular and diffuse reflections, and specular reflections tend to seriously affect the FPP accuracy. In order to analyze the negative impact of highly reflective surfaces on the FPP accuracy, highly reflective objects before and after spraying white powder are reconstructed. The grayscale curves and depth error curves for the same column of pixels in the highly reflective and nonhighly reflective fringe patterns are shown in Fig. 2(d). As can be seen, overexposure regions show significant depth errors due to luminance truncation, whereas other nonoverexposed regions also suffer from nonnegligible depth errors because the sinusoidal characteristics are destroyed and interfered.

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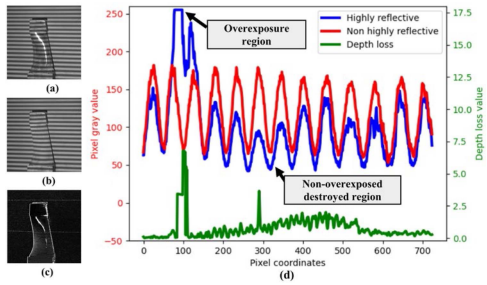


Fig. 2. (a) Highly reflective fringe. (b) Nonhighly reflective fringe. (c) Depth error map. (d) Sinusoidal features and depth error analysis of (a)–(c), with the left *y*-axis is pixel gray values for the highly reflective/nonhighly reflective fringes, and the right *y*-axis is depth error values.

III. Proposed Method

In order to propose a neural network suitable for the FPP high reflection removal task, a Y-FFC network which makes full use of the sinusoidal characteristics of fringe patterns is designed in this article. The high reflection removal task is essentially to separate diffuse and specular reflection components from the fringe pattern, so the imaging model adopted in our method [20] is expressed as follows:

$$I(x,y) = I_d(x,y) + I_s(x,y) \tag{2}$$

where I is the highly reflective fringe pattern, I_d is the diffuse reflection component, and I_s is the specular reflection component. I_d is the nonhighly reflective fringe pattern, and will serves as the final output of the Y-FFC.

A. Network Structure

The Y-FFC network, as shown in Fig. 3, consists of three cascaded subnetworks: an encoder network to extract the contextual information, a main decoder network to predict diffuse and specular reflection components, and a gradient decoder network to assist the main decoder network. The strategy of sharing encoder features between two decoders can reduce the complexity of the network model and improve the efficiency of information utilization [21].

The initial feature map is first extracted by a 3×3 convolution, then the initial feature map passes through five encoder layers, where the first two consist of a single convolution block (3×3 convolution + batch normalization + rectified linear unit) and a 2×2 max-pooling layer, and the last three encoder layers consist of two convolution blocks and a 2×2 max-pooling layer. In each encoder layer, the resolution of the feature map is halved, and the number of channels is doubled. The encoder is used to extract scene context information containing global structural and high-level semantic features. Subsequently, the feature maps output from the encoder are passed through an FFC [17], and then through a multiscale convolution block including $1 \times 1, 3 \times 3$, and 7×7 convolution kernels to collect information from different receptive fields. More details about FFC will be elaborated later.

After that, the output of the multiscale convolutional block is inputted to the main decoder, which utilizes the multiscale contextual information to predict the nonhighly reflective fringe pattern. The main decoder consists of five decoder layers that upsample the feature maps to their original size. Each decoder layer is a multibranch structure consisting of three 4×4 transposed convolutions. To fuse the encoder features into the decoder, attention blocks [22] are introduced to compensate for information loss caused by resampling. Compared to simple skip connections, the attention block can give different attention to different channels, realizing efficient information fusion. The output of the main decoder is the diffuse reflection, by which the specular reflection can be obtained with (2).

For sinusoidal fringe patterns, the gradient information can simplify the identification of highly reflective regions, because the grayscale gradient of overexposed regions changes from cosine wave to 0, whereas that in nonoverexposed destroyed regions becomes garbled. Based on the usefulness of the gradient information, a multiscale gradient decoder is designed with diffuse reflection gradients as output. The initial gradient features are extracted by 7×7 and 1×1 convolution, and then the gradient feature maps with different scales are obtained by four transposed convolution layers. These gradient feature maps are connected to the corresponding main decoder layers through pixel-wise addition, enabling the main decoder to focus more on high reflection regions with the help of gradient information.

B. Fast Fourier Convolution

Different from the input images of ordinary high reflection removal tasks, the FPP input is the sinusoidal fringe patterns, which are easier to represent in the frequency domain than in the spatial domain. To provide more effective theoretical support for the frequency domain method, both virtual data and real data with different disturbing factors are analyzed in Fig. 4. First, the column of pixels across the highly reflective region is taken in the sinusoidal fringe pattern, and the corresponding pixel gray curves are shown in Fig. 4. According to Fourier theory, any waveform can be decomposed into a weighted sum of sinusoids with different frequencies. Therefore, the Fourier transform (FT) is performed on these pixel gray curves to visualize its frequency components, as shown in the spectrum diagram in Fig. 4. It is worth noting that noise frequencies with amplitudes less than 5 are filtered out and only the positive frequency portion is shown for better visualization. It can be observed that the frequency components corresponding to the nonhighly reflective fringe patterns include the projection frequencies, the depth modulation frequencies, and the external interference frequencies. Whereas the highly reflective fringe patterns have more low and high frequency information, where the low frequencies are mainly due to the consistent luminance of the overexposed area, the high frequencies are introduced by the sudden change in luminance from the normal area to the overexposed area. Therefore, the highly reflective fringe patterns have more frequency components [23], which can be expressed as

$$I(x,y) = A(x,y) + \sum_{i=1}^{k} B_i(x,y)\cos\left[i\varphi(x,y)\right], k > 1 \tag{3}$$

where A is the background light intensity, B_i is the ith harmonic intensity, and φ is the phase. The separation of diffuse

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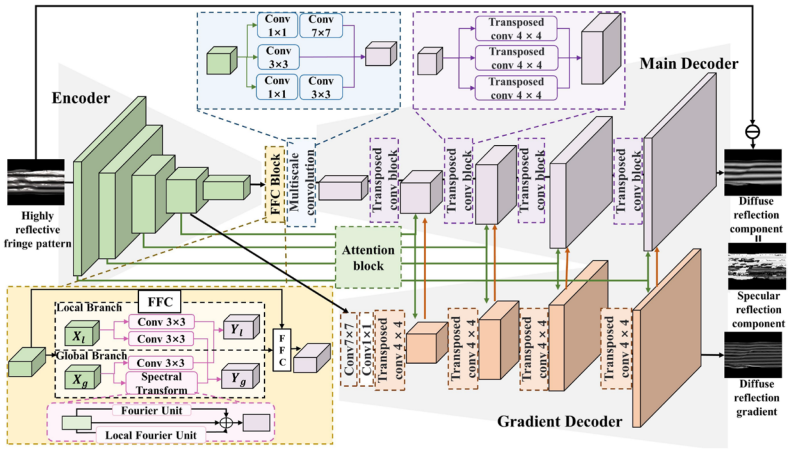


Fig. 3. Network structure of Y-FFC consists of three cascaded subnetworks: an encoder network for feature extraction, a main decoder network for obtaining diffuse and specular reflection components, and a gradient decoder network that assists the main decoder network.

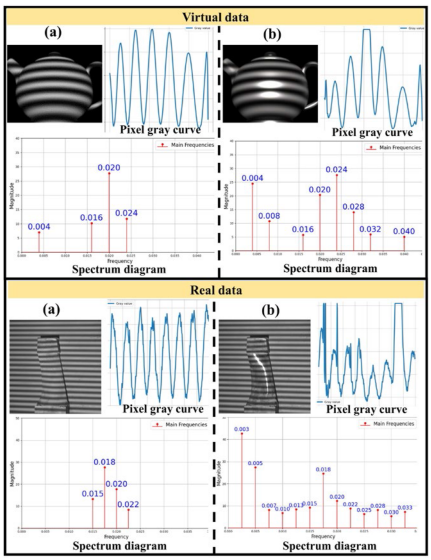


Fig. 4. Sinusoidal curves and the corresponding spectrum diagram of (a) nonhighly reflective and (b) highly reflective fringe patterns.

and specular reflection components is difficult to realize in the spatial domain. Fortunately, converting the fringe pattern to the frequency domain can simplify the weighted sum of multiple sinusoidal functions in (3) to a few constants, making the separation of diffuse and specular reflection components simpler. Thus, to facilitate feature processing in the frequency domain, this article introduces FFC.

The specific operation of FFC is shown in Fig. 3. The input features are divided in half on the channel dimension to obtain two subfeature maps: X_l and X_g. The subsequent path is divided into two branches, one of which only performs ordinary convolution, and since convolution is a local operation, this branch is called the local branch. The other branch converts the feature maps to the frequency domain before convolution, and since updating a single value in the frequency domain affects all the original features globally [24], this branch effectively realizes the global receptive field and is called the global branch. In the local branch, the output Y_l is obtained by adding X_l and X_g after 3×3 convolution respectively. In the global branch, the output Y_g is obtained by adding the feature map obtained by X_l after 3×3 convolution and the feature map obtained by X_g after spectral transform. The spectral transform consists of a Fourier unit (FU) and a local Fourier unit (LFU). The difference between the FU and the LFU is the size of the input feature maps, which is the entire input feature and 1/4 channel of the input feature, respectively. Both FU and LFU use FT to transform the original spatial features to the frequency domain, then perform 1×1 convolution, Batch Normalization, and rectified linear unit operations, and finally transform the data back to the spatial domain.

C. Loss Function

In order to ensure that the restored fringe pattern maintains correct sinusoidal characteristics, loss functions incorporating mean absolute error (MAE), structural similarity (SSIM) [25], and statistical loss (SL) [26] are designed. SSIM measures image similarity in terms of brightness, contrast, and structure, and SL focuses on the higher order SL. Hidden community detection [26] states that reducing the dominant structures can improve the detection of hidden structures. Therefore, the specular reflection component participates in the loss function calculation, which helps to extract diffuse reflections by flexibly changing the specular reflection weights. In addition, the diffuse reflection gradient output from the gradient decoder is also optimized by the loss function. The loss function is as follows:

$$L_d = \text{SSIM}(P_d - G_d) + \text{MAE}(P_d - G_d) + SL(P_d - G_d) \tag{4}$$

$$L_s = \text{SSIM}(P_s - G_s) + SL(P_s - G_s) \tag{5}$$

$$L_g = \text{SSIM}(P_g - G_g) \tag{6}$$

where L_d, L_s, and L_g represent the losses corresponding to diffuse reflection component, specular reflection component, and diffuse reflection gradient, respectively. P_d, P_s, and P_g represent their corresponding predicted values by our method,

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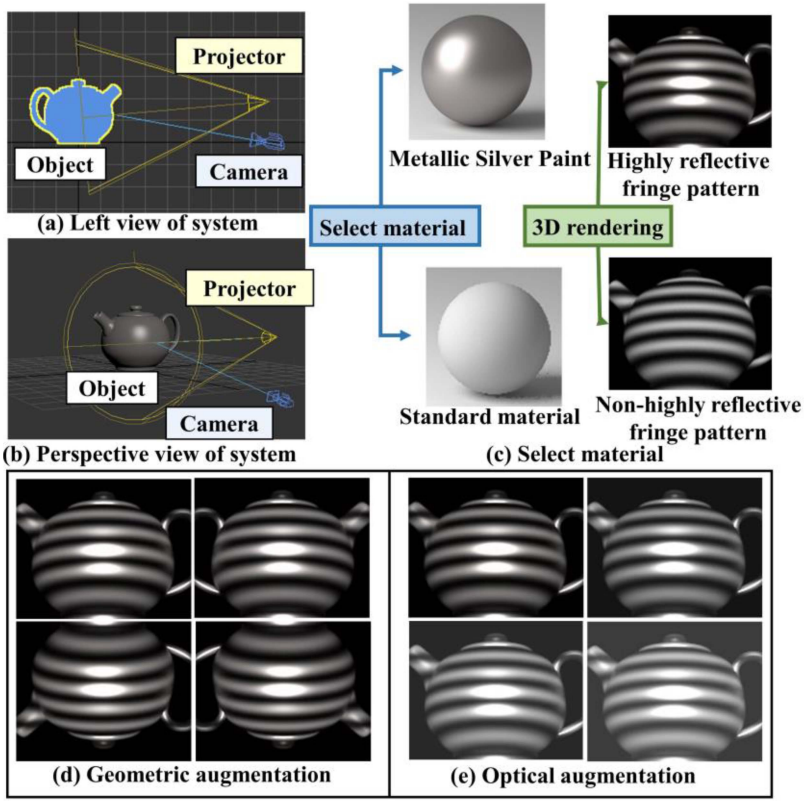


Fig. 5. Highly reflective fringe pattern dataset constructed by virtual software. (a) Left view of FPP system. (b) perspective view of FPP system. (c) Construction process. (d) and (e) Data augmentation.

and G_d, G_s, and G_g represent their corresponding ground truth values. The overall loss function is as follows:

$$\text{Loss} = L_d + L_s + L_g. \tag{7}$$

D. Dataset Preparation

The dataset constructed in this article is divided into two parts, a large-scale virtual dataset for network training and validation, and a real dataset constructed from industrial workpieces with complex shapes and varied reflectivity for testing experiments. Both virtual and real datasets should include highly reflective fringe pattern as Y-FFC inputs, and the nonhighly reflective fringe pattern as ground truth. The virtual dataset for training also needs to provide ground truth of specular reflection and diffuse reflection gradient. The specular reflection can be obtained by subtracting the diffuse reflection from the highly reflective fringe pattern, and the diffuse reflection gradient is calculated by using the Sobel operator on the diffuse reflection. These operations are simple and can be performed in batches without affecting the efficiency of dataset generation.

The virtual dataset is generated by constructing a digital twin of real FPP system using the virtual software 3Ds Max [11], as shown in Fig. 5. Models from the publicly available model library are selected as the virtual objects, and their reflective properties are adjusted by changing the surface material. Two sets of images are captured for each object, as shown in Fig. 5(c). The first set of images uses metallic silver paint on the surface, capturing the highly reflective fringe patterners. The second set uses standard materials that exhibit only diffuse reflections, capturing the nonhighly reflective fringe pattern.

To alleviate the lack of data diversity in the virtual dataset, both geometric augmentation and optical augmentation are used. Geometric augmentation consists of rotating and flipping the image, as shown in Fig. 5(d), and this augmentation enriches the position and orientation of objects in the image. Optical augmentation generates fringe patterns with different brightness by varying the exposure time, as shown in Fig. 5(e), and this augmentation ensures that the network performance is not affected by camera exposure parameters, ambient light intensity, and projector intensity.

IV. Experimentation and Analysis

Our network was developed by Pytorch and trained on an NVIDIA GeForce 3090 GPU. The network was trained 1000 epochs and took about 6 h. The batch size was 64 and the learning rate was 0.0001. The virtual dataset discussed in Section III-D contained 3600 data, of which 70% was used for training, 20% for validation, and 10% for testing. In this section, ablation experiments were first conducted to verify the enhancement of gradient and frequency information on the high reflection

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TABLE I
COMPARISON OF MSE AND SSIM OF FRINGE PATTERNS RESTORED BY Y-FFC, SUBNET1, AND SUBNET2

Methods	Training dataset		Test dataset		Validation dataset	
	MSE	SSIM	MSE	SSIM	MSE	SSIM
Subnet1	0.0097	0.9755	0.0166	0.9549	0.0102	0.9583
Subnet2	0.0014	0.9792	0.0020	0.9563	0.0019	0.9585
Y-FFC	**0.0004**	**0.9884**	**0.0009**	**0.9698**	**0.0009**	**0.9698**

removal effect. Then, the high reflection removal effect of Y-FFC was evaluated in the following aspects.

1) The high reflection removal effect of Y-FFC was directly evaluated based on the restored fringe patterns.
2) The phase retrieval results were used as metrics to evaluate the high reflection removal effect, and comparative experiments with the traditional and deep learning methods were carried out.
3) The 3D reconstruction results were used as metrics to evaluate the high reflection removal effect, and comparative experiments were also performed. It is worth mentioning that the test objects in the following experiments were typical highly reflective workpieces in industrial scenarios without participating in the network training, and their ground truth was obtained by spraying white powder on the surface.

A. Ablation Experiment

In order to validate the effectiveness of the FFC module and gradient decoder in the Y-FFC network, subnets 1 and 2 were designed to represent the networks after removing the FFC and the gradient decoder from the Y-FFC. In order to quantitatively compare the accuracies of Y-FFC and the two subnets on the training, validation, and test datasets, their MSE and SSIM are given in Table I. It can be seen that the training MSE of Y-FFC reaches 10^{-4} level (0.0004), the validation and test MSE is one orders of magnitude smaller than subnets, and the SSIM of Y-FFC is closer to 1.

In order to evaluate the effect in practical applications, the visualisation of the fringe patterns and the corresponding MSEs for the virtual and real scenes restored by Y-FFC and the two subnets are shown in Fig. 6. It can be seen that the fringe patterns restored by two subnets have obvious high reflection traces, whereas Y-FFC not only completely eliminates these traces, but also obtains smaller MSEs which are even an order of magnitude smaller than two subnets. Therefore, it can be concluded that Fourier convolution and gradient decoder can improve the high reflection removal performance.

B. Evaluation of High Reflection Removal Effect Based on Fringe Pattern

In this section, the high reflection removal effect of the Y-FFC network was evaluated by intuitive visual comparison, sine curve comparison, and quantitative comparison of the restored fringe patterns. The state-of-the-art high reflection removal networks high dynamic range fringe pattern improvement network (HDR-Net) [14] and dual channel unet and dilated residual networks

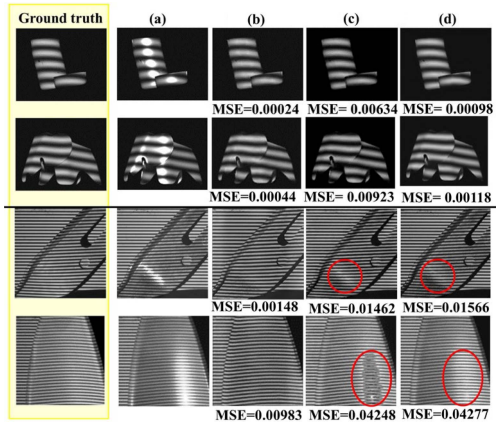


Fig. 6. Comparison of highly reflective fringe patterns before and after restoration using Y-FFC, Subnet1, and Subnet2. (a) High reflection. (b) Y-FFC. (c) Subnet1. (d) Subnet2.

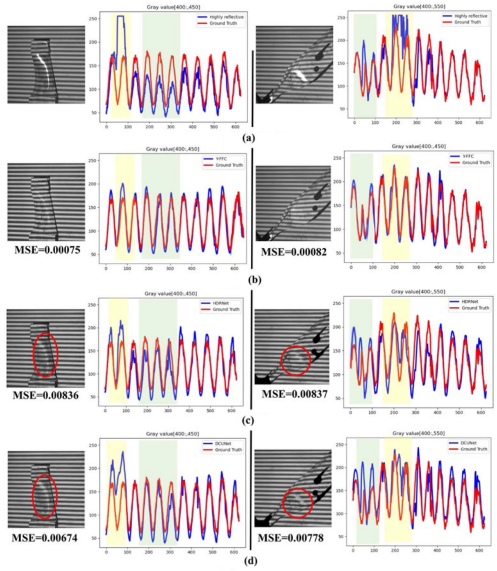


Fig. 7. Comparison of grayscale values of selected pixels column on fringe patterns before and after restoration using different methods. (a) Highly reflective. (b) Y-FFC. (c) HDR-Net. (d) DC-Unet.

(DC-Unet) [11] were selected as the comparison methods, and the results are shown in Fig. 7. As can be seen from the fringe patterns before and after restoration using the above methods and their corresponding MSEs in Fig. 7, Y-FFC can more completely remove the high reflection traces and the MSEs are an order of magnitude smaller than those of HDR-Net and DC-Unet. To verify the restoration effect of our method more intuitively, a column of pixels on the highly reflective fringe pattern, which passes through both overexposed and nonoverexposed regions, was randomly selected. The grayscale values of this column of pixels in the fringe patterns before and after restoration using above methods are also shown in Fig. 7. It can be seen that Y-FFC, HDR-Net, and DC-Unet can all solve the truncation problem caused by oversaturation of grayscale values, but the fringe patterns restored by DC-Unet lose sinusoidal characteristics, as shown in the yellow area in Fig. 7. In the nonoverexposed

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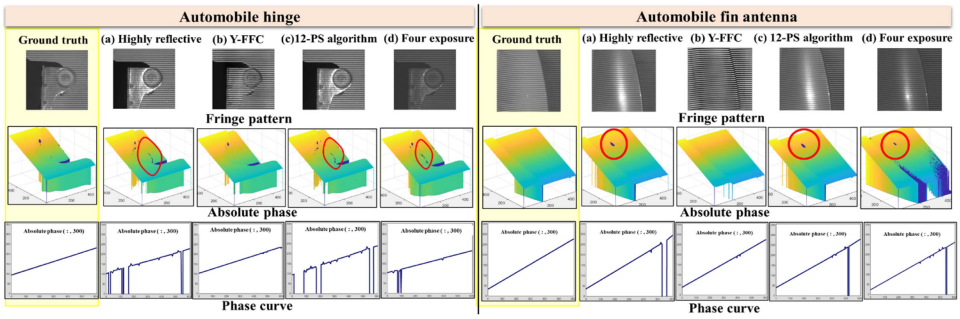


Fig. 8. Comparison of phase results of highly reflective fringe pattern before and after restoration using Y-FFC and traditional comparison methods.

TABLE II
COMPARISON OF MSE, SSIM, AND PSNR OF FRINGE PATTERNS RESTORED BY Y-FFC, HDR-NET, AND DC-UNET

Methods	Training_dataset			Test_dataset		
	MSE	SSIM	PSNR	MSE	SSIM	PSNR
HDR-Net	0.0006	0.9401	35.0178	0.0028	0.9075	26.8879
DC-Unet	0.0006	0.9364	35.2901	0.0016	0.9061	29.9348
Y-FFC	**0.0004**	**0.9884**	**36.8630**	**0.0009**	**0.9698**	**31.3840**
Methods	Validation_dataset			Real_dataset		
	MSE	SSIM	PSNR	MSE	SSIM	PSNR
HDR-Net	0.0021	0.8978	28.2701	0.0104	0.7552	16.8144
DC-Unet	0.0012	0.9264	30.8672	0.0107	0.7466	16.7791
Y-FFC	**0.0007**	**0.9670**	**31.7567**	**0.0012**	**0.8987**	**17.0808**

TABLE III
COMPARISON OF ABSOLUTE PHASE MAE BETWEEN Y-FFC, FOUR EXPOSURES METHOD, AND 12-STEP PS ALGORITHM

Methods	Automobile hinges	Automobile fin antenna
Highly reflective pattern	36.2202	23.2166
Four exposures method	35.7122	21.3737
12-step PS algorithm	7.1947	14.2747
Y-FFC	1.7903	0.9227

region, only the fringe patterns restored by Y-FFC have ideal sinusoidal characteristics, whereas the fringe patterns restored by HDR-Net and DC-Unet still have nonideal fluctuations, as shown in the green area in Fig. 7. In conclusion, our method not only accurately restores the overexposed regions, but also well preserves the fringe sinusoidal characteristic in nonoverexposed regions.

For quantitative evaluation, MSE, peak signal-to-noise ratio (PSNR), and SSIM were used to verify the performance of the above methods. As given in Table II, for training, testing, and validation losses, the Y-FFC achieves the MSE of 10^{-4} level, which is smaller than that of HDR-Net and DC-Unet. The SSIM of our method is larger than 0.96, which is closer to 1 than the other methods, and the PSNR is the largest among these methods. For practical scenarios, the MSE of our method can reach the 10^{-2} level, which is an order of magnitude smaller than the other two methods, and our method has the closest SSIM to 1 and the highest PSNR. In conclusion, Y-FFC can achieve the smallest error, indicating that our method outperforms the existing high reflection removal networks.

C. Evaluation of High Reflection Removal Effect Based on Phase Retrieval

All restored fringe patterns ultimately require phase retrieval for 3D reconstruction, so the phase retrieval of restored fringe patterns is important metrics to evaluate the effect of high reflection removal. In this section, the four-step PS algorithm and the 3-frequency heterodyne method were used for phase retrieval of the restored fringe patterns. The following are comparison experiments with traditional and deep learning methods.

1) Comparison of Y-FFC With Traditional Methods: The traditional methods of increasing phase shift steps [6] and multiple exposures method [5] are widely used in the field of high reflection removal due to their high accuracy and easy operation, so the 12-step PS algorithm and the Four exposures method were chosen as the comparative methods. It is worth noting that the 12-step PS algorithm can directly obtain the wrapped phase, whereas our method and the four exposures method can only restore the fringe patterns, and need to use the above algorithm for phase retrieval. In order to verify the practical application ability of our method in the manufacturing industry, the automobile hinge and fin antenna were selected as the test objects, and the absolute phase maps of above methods are shown in Fig. 8. For better presentation, a column of pixels which passes through the highly reflective region was randomly selected, and the corresponding absolute phase values of this pixel column are shown as phase curves in Fig. 8. As can be seen from Fig. 8, Y-FFC can obtain smoother and defect-free absolute phase maps, whereas the other two methods still have anomalies and invalid regions. More intuitively, the phase curves of our method are free of outliers and jumps, whereas the other two methods still have some jumps with phase plunges.

For quantitative evaluation, the absolute phase MAEs of our method and the above traditional methods are given in Table III. It can be seen that our method has the smallest error, with MAEs of 3.7903 and 0.9227 for the above two test objects, followed by the 12-step PS algorithm, and the worst result is obtained by the four-exposure method. In conclusion, even though the number of shots is several times higher than that of our method (12 x for 12-step PS method and 4 x for Four exposures method), the accuracy of the traditional methods is still lower than that of our method.

2) Comparison of Y-FFC With Deep Learning Methods: To highlight the superiority of Y-FFC in existing high reflection removal networks, the same phase retrieval experiments were

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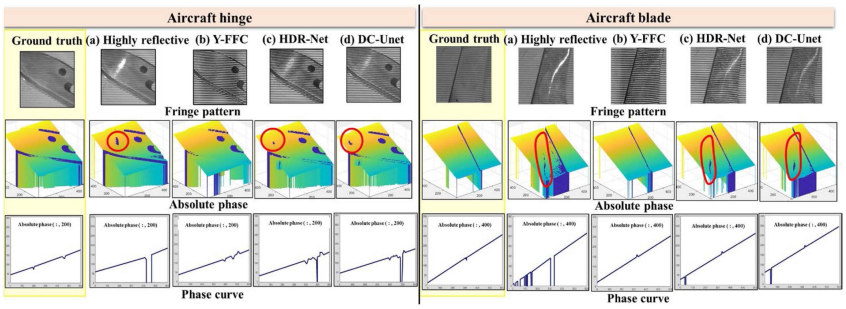


Fig. 9. Comparison of phase results of highly reflective fringe pattern before and after restoration using Y-FFC, HDR-Net, and DC-Unet.

TABLE IV
COMPARISON OF ABSOLUTE PHASE MAE BETWEEN Y-FFC, HDR-NET, AND DC-UNET

Input	Aircraft blades	Aircraft hinges
Highly reflective pattern	17.0508	41.3855
Restored by HDR-Net	2.4820	3.5104
Restored by DC-UNet	0.9572	0.9039
Restored by Y-FFC	0.3296	0.6413

used to compare our method with HDR-Net and DC-Unet. Moreover, to demonstrate the practicality of our method in the aerospace industry, the absolute phases of aircraft blades and hinges are shown in Fig. 9. It can be seen that our method obtains the smoothest phase maps without outliers and no jumps in phase curves, whereas the phase maps of HDR-Net and DC-Unet still have defective regions and some jumps in phase curves.

For quantitative evaluation, the absolute phase MAEs of our methods, HDR-Net, and DC-Unet are given in Table IV. It can be seen that on the above two aerospace workpieces, the Y-FFC is the most effective with MAEs of 0.3296 and 0.6413, followed by DC-Unet, and the worst is HDR-Net whose MAEs are one order of magnitude higher than Y-FFC. In summary, the above deep learning methods can effectively mitigate the negative impact of high reflections on the phase retrieval, but our method has higher accuracy on aerospace workpieces, implying that our method is more suitable for high reflection removal in industrial fields, such as aerospace.

D. Evaluation of High Reflection Removal Effect Based on 3D Reconstruction

The ultimate goal of high reflection removal is to obtain high-quality 3D reconstruction results, so the point cloud reconstruction results were used to evaluate the effect of high reflection removal in this experiment. This experiment compared our method with the 12-step PS algorithm, HDR-Net, and DC-Unet, where the 12-step PS algorithm obtained the wrapped phase directly, whereas the other methods firstly restored the fringe patterns, and then used the 4-step PS algorithm to calculate the wrapped phase. Then, the absolute phase was calculated using the 3-frequency heterodyne method and the point cloud was reconstructed using triangulation.

In the industry, the accuracy of the 3D shapes of highly reflective aircraft blades and hinges is crucial as it directly impacts aircraft safety, thus their point clouds obtained by the

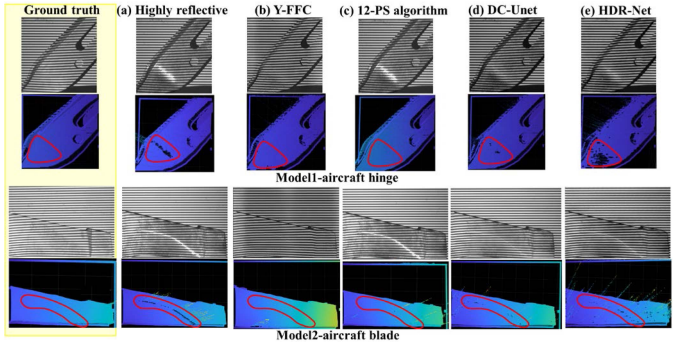


Fig. 10. Comparison of point clouds reconstructed from highly reflective fringe patterns before and after restoration using Y-FFC, 12-step PS algorithm, DC-Unet, and HDR-Net.

TABLE V
COMPARISON OF EPC AND DEPTH MAE BETWEEN Y-FFC, HDR-NET, DC-UNET, AND 12-STEP PS ALGORITHM

Methods	Aircraft hinge		Aircraft blades	
	EPC	depth MAE(mm)	EPC	depth MAE(mm)
Highly reflective	1465	0.0909	1497	0.1332
DC-UNet	188	0.0880	524	0.0977
HDR-Net	128	0.0828	102	0.0865
12-step PS	53	0.0037	95	0.0086
Y-FFC	0	0.0027	10	0.0041

above methods are shown in Fig. 10. As can be seen from the red-circled area in Fig. 10, the point clouds reconstructed directly from the highly reflective fringe patterns have massive outliers, whereas the Y-FFC method can effectively eliminate these outliers to generate smooth point clouds. The 12-step PS algorithm can also generate relatively smooth point clouds, but there are still some outliers in the point clouds of the aircraft blades. The point clouds of DC-Unet still suffer from some defects in the areas where high reflection traces are present in the restored fringe patterns. HDR-Net performs the worst, with defect points in both overexposed and nonoverexposed areas, indicating that the sinusoidal fringes in the nonoverexposed regions were disturbed during the fringe restoration process.

For quantitative evaluation, the 3D reconstruction accuracies of the above methods are given in Table V, and the evaluation index is the number of erroneous point clouds (EPC) that deviate from the ground truth. From Table V, it can be seen that compared to directly utilizing the highly reflective fringe patterns

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of these two test objects (with EPCs of 1465 and 1497), our method is almost free of errors (with EPCs of 0 and 10). Other comparison methods are able to partially eliminate outliers, but their EPCs are higher than our method. To state the 3D reconstruction accuracy more precisely, depth reconstruction was also performed, and the MAEs of the depth maps are given in Table V. It can be seen that our method obtains the smallest MAE, with values of 0.0027 and 0.0041 for the two test objects, which are an order of magnitude lower than those of HDR-Net and DC-Unet. In conclusion, our method accurately captures the shapes and details of aerospace workpieces, providing a feasible solution for 3D reconstruction of highly reflective workpieces in the industrial fields.

V. Conclusion

This article proposed a high reflection removal network, (Y-FFC, for the restoration of highly reflective fringe patterns. Y-FFC was pioneering because it not only solves the overexposure problem, but also attends to nonoverexposed regions where sinusoidal characteristics were also destroyed. Moreover, Y-FFC simplified the fringe restoration by fully utilizing the sinusoidal characteristics from the frequency and gradient domains, and large-scale virtual datasets and diverse loss functions were constructed to improve the network performance. Experimental results of high reflection removal using Y-FFC on industrial aircraft blades (30 mm depth) show that the predicted fringe patterns have a brightness MSE of 0.0043, a phase MAE of 0.3296, and a depth MAE of 0.0041 mm. In conclusion, Y-FFC provides an efficient and high-precision solution for 3D reconstruction of highly reflective surfaces in industrial scenes.

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